

AgentApp: Bridging Personalized and Collective Patterns for Mobile App Usage Prediction via Collaborative LLM Agents

Anonymous submission

Abstract

Mobile applications serve as essential gateways to daily services, making accurate usage prediction critical for enhancing service quality and responsiveness. In this paper, we propose AgentApp, a center-user multi-agent framework that bridges personalized and collective patterns via adaptive collaboration. Specifically, AgentApp employs a collective pattern builder on the cloud center-side to construct a behavioral knowledge base and retrieve shared usage patterns. On the local user-side, a personalized pattern profiler maintains long- and short-term memories of individual habits and intent, an app usage predictor performs context-aware reasoning by adaptively integrating collective and personalized insights, and an adaptive collaboration reflector refines the knowledge base and personalized memories while optimizing agent coordination. Evaluated on two real-world datasets, AgentApp consistently outperforms state-of-the-art baselines by over 7% across all metrics. Further analysis demonstrates the framework’s robustness and transferability, confirming that its collective behavioral knowledge base is both scalable and reusable across various models and geographic locations.

Introduction

Mobile app usage prediction aims to accurately forecast future user behavior by analyzing historical trajectories and environmental contexts (Shin, Hong, and Dey 2012; Huang et al. 2012; Parate et al. 2013). Deep learning-based methods (Sun et al. 2025; Zhang et al. 2023; Chen et al. 2019; Khaokaew et al. 2021; Xia et al. 2020; Huang et al. 2025) encode the intricate relationships between spatio-temporal contexts and underlying usage intentions. Recently, the exceptional semantic reasoning capabilities of Large Language Models (LLMs) (Lan et al. 2025; Kojima et al. 2022; Lin et al. 2024) have motivated researchers to leverage these models to either enrich session-level behaviors with semantic insights (Huang, Li, and Li 2025b) or serve as predictive engines (Khaokaew, Xue, and Salim 2024).

Despite remarkable advances, existing methods suffer from inherent structural limitations. As illustrated in Fig. 1, most previous approaches train a predictor over records from all users, which weakens deployment flexibility and may raise privacy concerns (Gan et al. 2023; Jiang et al. 2018; Hurtado, Raymond, and Soto 2021). Moreover, the knowledge learned during training is embedded in numerical latent weights, hindering the transferability and reuse of valuable

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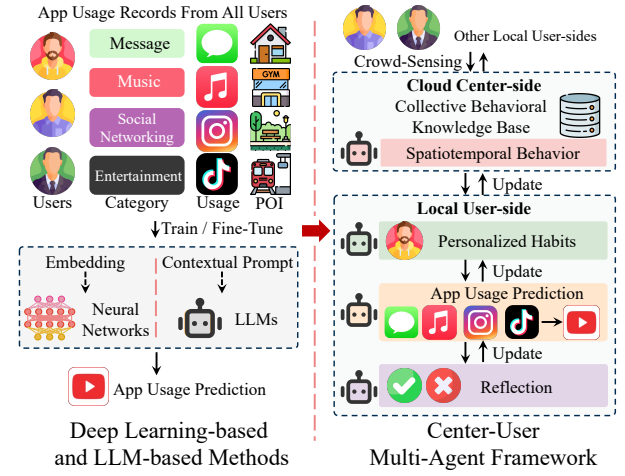


Figure 1: Previous approaches train a single predictor using aggregated data from all users, while center-user multi-agent framework maintains a collective behavioral knowledge base on the cloud that all users contribute to, and performs prediction on each user’s device.

behavioral insights (Xiong et al. 2024; Wang et al. 2023). In addition, the learned model usually remains static after deployment and cannot adjust its prediction strategy. In contrast, the center-user paradigm provides a flexible and privacy-preserving framework by separating cloud center-side behavioral knowledge construction from local user-side personalized prediction. Users collaboratively maintain a shared behavioral knowledge base, while local prediction retrieves only summarized behavioral knowledge without accessing other users’ records. This paradigm naturally fits a multi-agent framework, where the center-side focuses on collective knowledge construction and user-side independently performs reasoning. Through collaboration among agents, behavioral knowledge can be shared and refined as collective intelligence rather than being only absorbed into latent model weights.

In this paper, we aim to develop a center-user multi-agent framework for app usage prediction. Achieving this requires addressing the following three core challenges.

- **How to effectively extract and represent collective behavioral knowledge?** Mobile app usage exhibits inherent spatiotemporal patterns across the population, which is crucial for robust prediction. The challenge lies in enabling agents to extract generalizable insights from massive, cross-user datasets and structuring this ‘collective wisdom’ in an explicit, reusable format. Thus, agents can actively share and reference collective behavioral knowledge during inference.

- **How to coordinate personalized user intent with collective behavioral patterns?** App usage is also strongly shaped by individual habits, short-term shifts in intent, and personal preferences. However, collective pattern and personalized habits operate at different granularities and can occasionally conflict. Therefore, the second challenge is to orchestrate collaboration between agents analyzing collective patterns and those focusing on personalized signals.

- **How to establish an adaptive collaboration mechanism for robust prediction?** In a multi-agent framework, predefined coordination schemes often fail to capture the varying reliability of specialized agents across scenarios. The challenge is to enable agents to dynamically adjust their influence and coordination strategies. Therefore the framework can resolve ambiguities during prediction while evolving to deliver increasingly robust performance over time.

To address the challenges, we propose AgentApp, which consists of a cloud center-side and local user-sides. On the **cloud center-side**, a collective pattern builder maintains a collective behavioral knowledge base of shared app usage behavioral patterns. On each **local user-side**, a personalized pattern profiler stores user memories of personalized usage habits and recent intent; an app usage predictor coordinates collective and personalized patterns to predict the next app; and an adaptive collaboration reflector evaluates prediction performance and dynamically adapts the collaboration within the framework, enabling self-evolution over time. Specifically, on the cloud center-side, the collective pattern builder extracts behavioral patterns from crowd-sensed data and transforms them into natural-language descriptions. By converting raw data into shared, high-level insights, this process addresses the **first** challenge. In parallel, the personalized pattern profiler maintains long- and short-term memories of individual users to summarize their preferences and dynamic intent. The app usage predictor then employs an adaptive collaboration mechanism to jointly consider the collective patterns with personalized characteristics and infer the next app, addressing the **second** challenge. The adaptive collaboration reflector generates reflection signals to other agents based on their past performance after each prediction. By refining semantic descriptions in the collective knowledge base, optimizing how the profiler captures user memories, and modulating agent coordination strategies, the system ensures robust predictions through automated updates, addressing the **third** challenge.

Our main contributions can be summarized as follows:

- We investigate the use of multi-agent frameworks for predicting mobile app usage.
- We propose AgentApp, which consists of Collective Pattern Builder, the Personalized Pattern Profiler, the App Usage Predictor, and the Adaptive Collaboration Reflector.

- Experiments on two real-world datasets shown that AgentApp consistently outperforms baselines by over 7%. Further analyses verify the robustness and transferability.

Related Work

App Usage Prediction

Existing app usage prediction methods model sequential dependencies and contextual behavioral patterns. Statistical and neural models learn app transition regularities from historical records (Natarajan, Shin, and Dhillon 2013; Katsarou, Yu, and Beierle 2022; Xu et al. 2018; Cheng et al. 2023), while context-aware approaches incorporate temporal, spatial, semantic, and user-specific information (Huang, Li, and Li 2025a; Liao et al. 2013; Lee, Cho, and Choi 2019; Ouyang et al. 2022; Wang et al. 2021; Zhao et al. 2019; Khaokaew et al. 2021; Huang et al. 2025). More recently, LLM-based methods enhance behavioral representation and prediction through semantic modeling and contextual reasoning (Huang, Li, and Li 2025b; Khaokaew, Xue, and Salim 2024; Lan et al. 2025). However, these methods generally integrate personalized and cross-user information within a unified predictor, leaving collective knowledge implicit and lacking adaptive refinement of behavioral signals.

Multi-Agent Frameworks

Single-agent systems decompose complex tasks through step-wise reasoning, prompting, and tool invocation (Durante et al. 2024; Gao et al. 2024; Beneduce, Lepri, and Luca 2025), while multi-agent frameworks assign specialized roles and exchange intermediate results for collaborative reasoning (Calderón et al. 2025; Wang et al. 2024a, 2025; Jiang et al. 2023). Such frameworks support modular task decomposition, knowledge reuse, and flexible orchestration (Fang et al. 2023; Zhang et al. 2025), and have been applied to trajectory prediction (Feng et al. 2025; Ju et al. 2025), human behavior simulation (Fatemi, Halcrow, and Perozzi 2024; Wang et al. 2024b), and question answering (Liu et al. 2023; Ding et al. 2024). Their potential for app usage prediction remains underexplored, particularly for coordinating collective and personalized patterns and adapting their collaboration through prediction feedback.

Preliminaries

Mobile App Usage Behavior

App usage behavior consists of a series of events reflecting when, where, and which apps a user engages with. These behaviors are influenced by daily routines and functional requirements under varying spatiotemporal conditions. Prior studies (Li et al. 2021; Wang et al. 2019) have shown that users under similar spatiotemporal contexts often exhibit recurring regularities and shared app usage patterns. For example, users following similar daily routines may engage in fitness activities accompanied by music apps in the morning and use entertainment apps during evening commutes. Despite variations in specific locations or timestamps, their behaviors may share similar activity transitions and dominant app categories, forming a common “leisure-oriented” pattern. In contrast, users following work-centered routines may

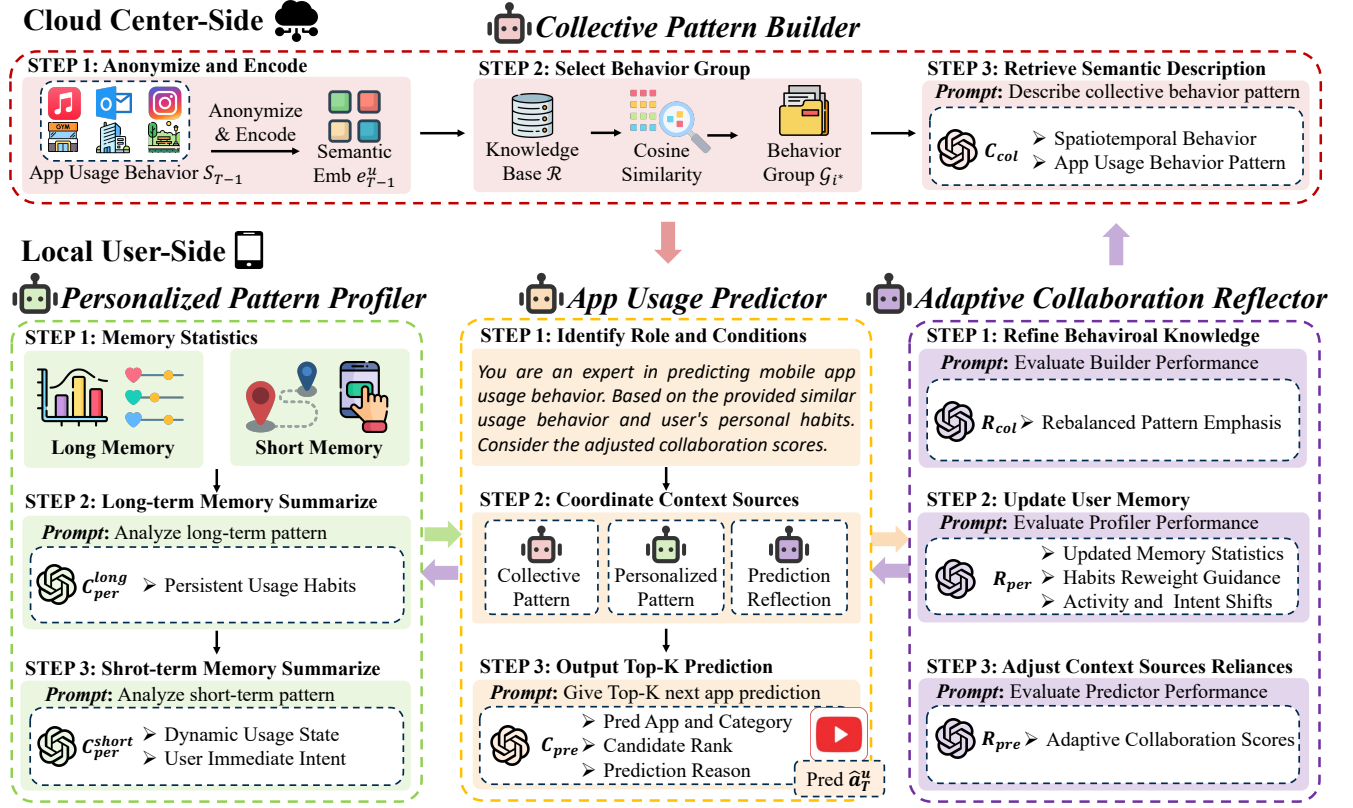


Figure 2: Framework overview of AgentApp, which consists of four collaborative agents: a collective pattern builder, a personalized pattern profiler, an app usage predictor, and an adaptive collaboration reflector.

exhibit distinct transitions, functional demands, and app preferences, thereby forming a “work-oriented” pattern. These observations suggest that app usage behaviors under similar contexts tend to converge toward common behavioral archetypes, whereas different contexts produce distinct behavioral signatures. Modeling such collective regularities can therefore provide behavioral insights beyond individual-level variations and support more accurate app usage prediction.

Problem Definition

Given historical app usage behaviors and contextual information, the goal of mobile app usage prediction is to estimate which app a user will launch next. Mathematically, the mobile app usage behavior of a user u is represented as:

$$S_T^u = \{s_1^u, s_2^u, \dots, s_T^u\}, \quad (1)$$

where T denotes the sequence length of app usage behavior, and each app usage event s_i^u is defined as:

$$s_i^u = (l_i^u, t_i^u, a_i^u), \quad (2)$$

where l_i^u denotes the location, t_i^u denotes the timestamp, and a_i^u denotes the used app. Given the historical app usage behavior S_{T-1}^u , and the current spatiotemporal context (l_T^u, t_T^u) , the objective is to design a model $f(\cdot)$ that predicts the next app usage $\hat{a}_T^u$:

$$\hat{a}_T^u = f(S_{T-1}^u, l_T^u, t_T^u). \quad (3)$$

AgentApp

Framework Overview

The overall framework of AgentApp is illustrated in Fig. 2. AgentApp consists of a cloud center-side and local user-sides. On the **cloud center-side**, a collective pattern builder maintains a collective behavioral knowledge base where human app usage behaviors are organized in groups to represent shared collective behavioral patterns. On each **local user-side**, a personalized pattern profiler provides personalized patterns by capturing long-term usage preferences and short-term intent shifts; an app usage predictor integrates collective behavioral patterns and personalized user intents and performs adaptive reasoning to predict which app the user will open next; and an adaptive collaboration reflector evaluates performance of agents and generates reflection signals, refining agents’ reasoning processes and dynamically adjusting their coordination.

Collective Pattern Builder

On the cloud center-side, the collective pattern builder constructs a collective behavioral knowledge base and distills semantic descriptions of shared app usage patterns. After crowd-sensing anonymized records across users, the builder clusters semantically similar behaviors into distinct pattern groups and generates textual summaries. During inference, the builder compares the observed behavior with patterns

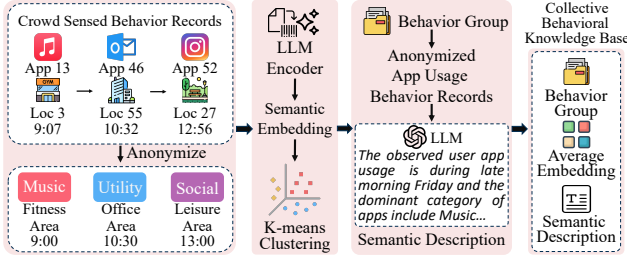


Figure 3: Pipeline of constructing the collective behavioral knowledge base, including crowd-sensing and anonymization, semantic encoding and embedding clustering, and pattern description generation.

in the knowledge base, and delivers collective behavioral insights to the app usage predictor on the local user-side. Furthermore, reflection signals from the adaptive collaboration reflector guide the builder in refining or expanding the knowledge base as new data arrives. This design enables AgentApp to leverage collective behavioral regularities while maintaining an evolving repository of prior knowledge.

Knowledge Base Construction The collective behavioral knowledge base $\mathcal{R}$ organizes app usage behaviors and enables the retrieval of group-level behavioral contexts relevant to the given usage behavior S_{T-1}^u for downstream prediction. The pipeline of knowledge base construction is demonstrated in Fig. 3. To preserve privacy during crowd-sensing, before reporting to the cloud center-side, the records of each user u' are anonymized, including mapping exact locations to regional Point-Of-Interest (POI) types, precise timestamps to discrete intervals defined by a fixed unit time window ΔT_{disc} , and specific app IDs to broader app categories. After receiving app usage behavior, an LLM encoder transforms each behavior into a semantic embedding $e_{T'}^{u'}$ and the builder employs K-means clustering to identify K_{group} distinct clusters. This classifies app usage behaviors into K_{group} groups, where behaviors within each group share common patterns. For each behavior group $\mathcal{G}_i$, we calculate the mean embedding $\bar{e}_i$ and propose a center-based sampling strategy where a predefined center ratio R_{center} guides the builder to select N_{seq} usage behaviors that cover both core behavioral patterns and significant variations based on cosine similarity. These representative behaviors are then provided to an LLM to generate a natural-language description, Desc_i . The collective behavioral knowledge base is defined as:

$$\mathcal{R} = \{(\mathcal{G}_i, \bar{e}_i, \text{Desc}_i)\}_{i=1}^{K_{\text{group}}}. \quad (4)$$

Furthermore, guided by reflection signals, the builder continuously refines or expands $\mathcal{R}$ through ongoing crowd-sensing.

Collective Pattern Retrieval Upon establishing the collective behavioral knowledge base $\mathcal{R}$, the builder identifies the group whose behavioral pattern most closely matches the observed usage behavior S_{T-1}^u , retrieves the corresponding semantic description, and sends it to the local user-side. Specifically, the builder first encodes the observed behavior S_{T-1}^u into semantic embedding e_{T-1}^u and then computes

the pairwise cosine similarity between e_{T-1}^u and each cluster centroid $\bar{e}_i$, selecting the i^* -th cluster with the highest similarity as the most relevant match:

$$i^* = \arg \max_{1 \leq i \leq K_{\text{group}}} \frac{\langle e_{T-1}^u, \bar{e}_i \rangle}{\|e_{T-1}^u\| \|\bar{e}_i\|}. \quad (5)$$

The behaviors within $\mathcal{G}_{i^*}$ represent the collective patterns most similar to the target user's recent activity. The collective pattern builder then provides the semantic description Desc_{i^*} as the collective context C_{col} to the app usage predictor on the local user-side. This context C_{col} summarizes category-level transitions of similar behavioral archetypes. By providing population-level references, the builder supplies informative prior knowledge of collective patterns to enhance the final reasoning process.

Personalized Pattern Profiler

On each local user-side, a personalized pattern profiler maintains long- and short-term user memory, capturing app usage preferences, personalized intent, and recent behavioral transitions. This profiling enables the app usage predictor to incorporate customized behavioral patterns in predictions. The long-term memory tracks persistent usage habits such as usage frequency of categories or specific apps across various locations and times. Memory size M_{size} determines the number of retained categories and apps. Given the spatiotemporal context (l_T^u, t_T^u) at prediction step T , the profiler retrieves the corresponding long-term memory statistics $\mathcal{M}_{\text{long}}^u(l_T^u, t_T^u)$. Conversely, the short-term memory records the observed app usage behavior S_{T-1}^u , capturing dynamic usage states, the user's immediate intent, and behavioral shifts.

The profiler leverages an LLM to distill memories into natural-language summaries. The LLM processes structured long-term memory statistics and short-term memory into concise summaries $C_{\text{per}}^{\text{long}}$ and $C_{\text{per}}^{\text{short}}$, which serve as customized input for the app usage predictor and consist the personalized pattern context C_{per} . Furthermore, guided by reflection signals, the profiler continuously refines memory based on newly observed behaviors. This iterative update mechanism ensures increasingly accurate and personalized contextual summarizations for future predictions.

App Usage Predictor

On each local user-side, an app usage predictor integrates collective behavioral patterns and personalized user intent to generate predictions and interpretable reasons. To enhance robustness across diverse scenarios, the predictor utilizes reflection signals from the adaptive collaboration reflector to dynamically calibrate reliance on multiple contextual sources. During the inference phase, the predictor jointly analyzes the collective pattern knowledge C_{col} and the personalized pattern context C_{per} . We employ an adaptive collaboration score set to maintain the reflection signal R_{app} . These scores are updated after each prediction and quantify the appropriate weight for each input source: higher scores grant a context greater influence over the final reasoning, while lower scores down-weight less reliable information. Guided by the adaptive collaboration mechanism, the predictor synthesizes contextual insights from other agents to

estimate the user’s next application. The predictor generates a prediction result C_{app} , which comprises a ranked list of the top- K candidate apps $\hat{a}_T^{u,(j)}$ alongside their corresponding interpretable reasoning contexts $r_T^{(j)}$:

$$C_{\text{app}} = \{(\hat{a}_T^{u,(j)}, r_T^{(j)})\}_{j=1}^K. \quad (6)$$

By jointly reasoning over collective behavioral regularities and personalized patterns, the predictor captures both persistent long-term habits and transient short-term intent. The adaptive collaboration mechanism dynamically calibrates the coordination, ensuring the final prediction effectively balances population-level trends with individual preferences.

Adaptive Collaboration Reflector

On each local user-side, an adaptive collaboration reflector coordinates agent interactions based on historical prediction performance. After each prediction, the reflector leverages an LLM to evaluate the contributions of participating agents and infer reflection signals to refine the collective knowledge base, update personalized memory summaries, and calibrate reliance on various contextual sources.

Collective Pattern Reflection Collective pattern reflection aims to refine the semantic descriptions within $\mathcal{R}$, thereby enhancing the quality of collective contexts for future predictions. Specifically, the reflector utilizes the LLM to assess the contribution of the collective context C_{col} to the final result and sends back to the cloud center-side a reflection signal R_{col} that guides the builder to rebalance the pattern’s emphasis by reinforcing informative spatiotemporal regularities and salient app categories. Based on R_{col} , the builder regenerates an updated description Desc_{i*}' .

Personalized Pattern Reflection Personalized pattern reflection optimizes how the profiler summarizes user memory based on prior performance. After each prediction, the profiler updates the long- and short-term memories with the ground-truth a_T^u . Simultaneously, the reflector evaluates the effectiveness of the personalized contexts C_{per} and generates reflection signals R_{per} . The long-term reflection guides the LLM to reweight persistent app categories and habits, while short-term reflection emphasizes recent activity shifts and rapid intent transitions. The profiler iteratively recalibrates its summarization strategy, producing personalized representations that remain consistent with evolving user behavior.

App Usage Prediction Reflection App usage prediction reflections facilitate dynamic adjustment of agent collaboration during the reasoning process. By assessing the contribution of disparate contextual sources, the reflector enables the predictor to adaptively modulate reliance on each input under varying scenarios. Specifically, the reflector collects the prediction result C_{app} , the ground-truth a_T^u , and contextual signals from different agents, and prompts an LLM to infer a collaboration signal R_{app} , which contains adaptive reliance scores for the predictor to modulate the importance of different contextual sources.

Experiments

In this section, we conduct extensive experiments on two real-world datasets of mobile app usage to evaluate the performance of AgentApp.

Experimental Settings

Dataset We evaluate the performance of AgentApp on Shanghai and Nanchang mobile app usage datasets, which were collected by major network operators over a one-week period, covering the entire metropolitan regions. Each record corresponds to an app usage event and consists of an anonymized user identifier, a timestamp, a location ID, the accessed app, and app category. The appendix section *Details of Datasets* provides detailed descriptions and statistics for both datasets.

Baselines and Experimental Settings We compare AgentApp with SOTA baselines to evaluate the performance on app usage prediction, including the deep learning method AppUsage2Vec (Zhao et al. 2019), context-aware deep learning approaches CoSEM (Khaokaew et al. 2021) and CADHG (Huang et al. 2025), as well as LLM-based methods SMAUG (Huang, Li, and Li 2025b), MAPLE (Khaokaew, Xue, and Salim 2024), and PIGEON (Lan et al. 2025). Detailed baseline descriptions are provided in the appendix section *Details of Baselines*. All baselines use the same LLM backbone and experimental settings, and are allowed the on-line access. We employ Accuracy at K (ACC@K), Mean Reciprocal Rank at K (MRR@K), and Normalized Discounted Cumulative Gain at K (NDCG@K) to evaluate the performance. Metric definitions and experimental implementation details are provided in the appendix sections *Details of Metrics* and *Details of Implementation*, respectively.

Experimental Performance

Overall Performance Table 1 reports the overall app usage prediction performance of AgentApp and SOTA baselines. Overall, AgentApp achieves consistent improvements across both datasets. Compared with SMAUG, AgentApp further improves app prediction MRR@3 by 12.40% on Nanchang, benefiting from the reflection-driven adaptive collaboration mechanism that continuously refines and coordinates contextual information instead of keeping enhanced knowledge fixed during inference. Compared with MAPLE, AgentApp achieves ACC@1 improvements of 12.94% on Nanchang by replacing single black-box LLM inference with structured multi-agent reasoning that integrates multiple contextual sources in a controllable and interpretable manner. Besides, AgentApp surpasses PIGEON, which combines GCN-based collective behavioral knowledge with an LLM, improving ACC@5 by 16.72% on Nanchang. This gain mainly comes from the stronger semantic representation and reasoning capability of LLMs than low-dimensional graph embeddings.

Ablation Analysis We conduct an ablation study by comparing the AgentApp against four variants: **w/o Collective** (removes the collective pattern builder), **w/o Long-term** (removes inherent habits), **w/o Short-term** (removes dynamic

Table 1: Overall app and app category prediction performance on the Shanghai and Nanchang datasets. Results are reported as mean $\pm$ standard deviation over five independent runs. Bold and underlined values denote the best and second-best.

Methods	App Prediction							App Category Prediction			
	ACC@1	ACC@3	ACC@5	MRR@3	MRR@5	NDCG@3	NDCG@5	ACC@1	ACC@3	MRR@3	NDCG@3
Shanghai											
APPU2Vec	0.3479 $\pm$ 0.0035	0.5556 $\pm$ 0.0023	0.6767 $\pm$ 0.0014	0.4427 $\pm$ 0.0010	0.4654 $\pm$ 0.0009	0.4865 $\pm$ 0.0015	0.5247 $\pm$ 0.0011	0.5335 $\pm$ 0.0043	0.6754 $\pm$ 0.0018	0.5944 $\pm$ 0.0009	0.6153 $\pm$ 0.0013
CoSEM	0.3444 $\pm$ 0.0026	0.5769 $\pm$ 0.0016	0.6842 $\pm$ 0.0011	0.4575 $\pm$ 0.0016	0.4751 $\pm$ 0.0021	0.4933 $\pm$ 0.0008	0.5249 $\pm$ 0.0020	0.5284 $\pm$ 0.0035	0.6764 $\pm$ 0.0012	0.6002 $\pm$ 0.0010	0.6032 $\pm$ 0.0014
CADHG	0.3485 $\pm$ 0.0029	0.5690 $\pm$ 0.0014	0.6680 $\pm$ 0.0019	0.4427 $\pm$ 0.0019	0.4654 $\pm$ 0.0016	0.4747 $\pm$ 0.0013	0.5161 $\pm$ 0.0018	0.5148 $\pm$ 0.0046	0.6639 $\pm$ 0.0020	0.6121 $\pm$ 0.0014	0.6283 $\pm$ 0.0014
SMAUG	0.3799 $\pm$ 0.0032	0.5987 $\pm$ 0.0008	0.7159 $\pm$ 0.0008	0.4556 $\pm$ 0.0019	0.5127 $\pm$ 0.0019	0.5153 $\pm$ 0.0009	0.5637 $\pm$ 0.0019	0.5336 $\pm$ 0.0046	0.7698 $\pm$ 0.0008	0.6384 $\pm$ 0.0022	0.6723 $\pm$ 0.0022
MAPLE	0.4103 $\pm$ 0.0045	0.6465 $\pm$ 0.0022	0.7229 $\pm$ 0.0021	0.4737 $\pm$ 0.0018	0.5266 $\pm$ 0.0018	0.5382 $\pm$ 0.0022	0.5634 $\pm$ 0.0023	0.5523 $\pm$ 0.0039	0.7895 $\pm$ 0.0013	0.6367 $\pm$ 0.0009	0.6784 $\pm$ 0.0009
PIGEON	0.3812 $\pm$ 0.0028	0.5916 $\pm$ 0.0009	0.7302 $\pm$ 0.0018	0.4665 $\pm$ 0.0018	0.4802 $\pm$ 0.0014	0.4912 $\pm$ 0.0009	0.5338 $\pm$ 0.0022	0.5271 $\pm$ 0.0033	0.6407 $\pm$ 0.0012	0.6162 $\pm$ 0.0014	0.6026 $\pm$ 0.0023
AgentApp	0.4502 $\pm$ 0.0043	0.7126 $\pm$ 0.0023	0.7824 $\pm$ 0.0019	0.5595 $\pm$ 0.0018	0.5763 $\pm$ 0.0015	0.5964 $\pm$ 0.0023	0.6252 $\pm$ 0.0021	0.6051 $\pm$ 0.0037	0.8209 $\pm$ 0.0021	0.6583 $\pm$ 0.0019	0.6904 $\pm$ 0.0013
Improvement	+9.72%	+10.22%	+7.15%	+18.11%	+9.44%	+10.81%	+10.91%	+9.56%	+3.98%	+3.12%	+1.77%
Nanchang											
APPU2Vec	0.2303 $\pm$ 0.0037	0.4022 $\pm$ 0.0015	0.4685 $\pm$ 0.0017	0.3197 $\pm$ 0.0021	0.3363 $\pm$ 0.0019	0.3436 $\pm$ 0.0013	0.3715 $\pm$ 0.0012	0.2992 $\pm$ 0.0034	0.5948 $\pm$ 0.0010	0.4273 $\pm$ 0.0014	0.4705 $\pm$ 0.0009
CoSEM	0.2518 $\pm$ 0.0033	0.4382 $\pm$ 0.0016	0.4768 $\pm$ 0.0015	0.3482 $\pm$ 0.0019	0.3579 $\pm$ 0.0013	0.3410 $\pm$ 0.0010	0.3874 $\pm$ 0.0012	0.3140 $\pm$ 0.0041	0.5713 $\pm$ 0.0014	0.3928 $\pm$ 0.0019	0.4390 $\pm$ 0.0011
CADHG	0.2363 $\pm$ 0.0040	0.4374 $\pm$ 0.0022	0.5591 $\pm$ 0.0012	0.3240 $\pm$ 0.0017	0.3517 $\pm$ 0.0016	0.3526 $\pm$ 0.0010	0.4029 $\pm$ 0.0018	0.3179 $\pm$ 0.0032	0.5870 $\pm$ 0.0017	0.4338 $\pm$ 0.0009	0.4741 $\pm$ 0.0010
SMAUG	0.2649 $\pm$ 0.0027	0.4137 $\pm$ 0.0010	0.5681 $\pm$ 0.0023	0.3424 $\pm$ 0.0017	0.3573 $\pm$ 0.0017	0.3691 $\pm$ 0.0022	0.3986 $\pm$ 0.0012	0.3332 $\pm$ 0.0043	0.5749 $\pm$ 0.0012	0.4781 $\pm$ 0.0009	0.5286 $\pm$ 0.0016
MAPLE	0.2762 $\pm$ 0.0040	0.4582 $\pm$ 0.0014	0.5550 $\pm$ 0.0009	0.3545 $\pm$ 0.0019	0.3769 $\pm$ 0.0013	0.3816 $\pm$ 0.0019	0.4216 $\pm$ 0.0011	0.3021 $\pm$ 0.0038	0.5914 $\pm$ 0.0009	0.4652 $\pm$ 0.0017	0.5196 $\pm$ 0.0017
PIGEON	0.2719 $\pm$ 0.0025	0.4021 $\pm$ 0.0018	0.5449 $\pm$ 0.0021	0.3281 $\pm$ 0.0009	0.3600 $\pm$ 0.0009	0.3767 $\pm$ 0.0018	0.4068 $\pm$ 0.0022	0.2839 $\pm$ 0.0040	0.5619 $\pm$ 0.0013	0.4505 $\pm$ 0.0021	0.4735 $\pm$ 0.0023
AgentApp	0.3113 $\pm$ 0.0040	0.5190 $\pm$ 0.0023	0.6360 $\pm$ 0.0019	0.3850 $\pm$ 0.0013	0.4121 $\pm$ 0.0023	0.4291 $\pm$ 0.0017	0.4678 $\pm$ 0.0009	0.3678 $\pm$ 0.0034	0.6468 $\pm$ 0.0017	0.4962 $\pm$ 0.0021	0.5451 $\pm$ 0.0010
Improvement	+12.71%	+13.27%	+11.95%	+8.60%	+9.34%	+12.45%	+10.96%	+10.38%	+8.74%	+3.79%	+3.12%

Table 2: Ablation analysis of AgentApp on the Shanghai and Nanchang datasets. Bold indicates the best result.

Variants	App Prediction			
	ACC@1	ACC@3	MRR@5	NDCG@5
Shanghai				
w/o Collective	0.4027	0.6710	0.5362	0.5530
w/o Long-term	0.3890	0.6404	0.4924	0.5553
w/o Short-term	0.4214	0.6953	0.5418	0.5850
w/o Reflector	0.3988	0.6753	0.5233	0.5738
AgentApp	0.4505	0.7130	0.5760	0.6254
Nanchang				
w/o Collective	0.2816	0.4859	0.3890	0.4441
w/o Long-term	0.2650	0.4650	0.3663	0.4211
w/o Short-term	0.2984	0.4943	0.3939	0.4450
w/o Reflector	0.2799	0.4840	0.3795	0.4309
AgentApp	0.3116	0.5189	0.4119	0.4677

intent), and **w/o Reflector** (turns off the adaptive collaboration mechanism). As shown in Table 2, removing any major component degrades performance. Among the variants, w/o Long-term causes the largest drop, indicating that persistent usage habits provide the most fundamental signal for app prediction. w/o Collective and w/o Reflector also lead to clear performance declines, demonstrating the importance of cross-user behavioral regularities and adaptive coordination.

Hyperparameter Analysis We further conduct hyperparameter analysis for the collective pattern builder. K_{group} controls the number of behavior pattern groups, while ΔT_{disc} determines the temporal granularity. As shown in Fig. 4, performance first improves and then saturates or declines as, indicating that overly coarse settings fail to distinguish behavioral patterns, whereas overly fine settings may fragment temporal regularities. Based on the results, we set $K_{\text{group}} = 100$ and $\Delta T_{\text{disc}} = 30$ minutes as the default configuration. Additional analyses of the number of sampled representative sequences N_{seq} , the center sampling ratio R_{center} , the memory size M_{size} , and the historical sequence length T are provided in the appendix section *Hyperparameter Analysis*.

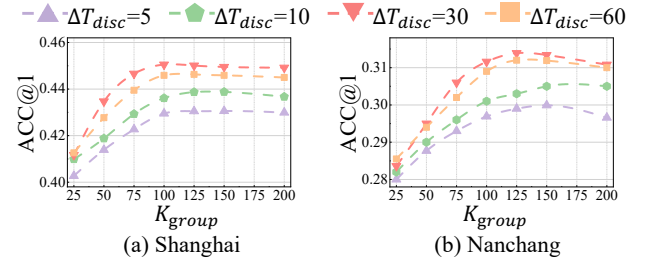


Figure 4: Sensitivity of AgentApp to the number of pattern groups K_{group} and the temporal discretization window ΔT_{disc} (unit: minute) on the Shanghai and Nanchang datasets.

Collective Knowledge Analysis We analyze the characteristics of the behavioral pattern groups within $\mathcal{R}$. Appendix section *Collective Knowledge Analysis* demonstrates that behavioral pattern groups in both Shanghai and Nanchang exhibit long-tailed popularity distributions, suggesting that the knowledge base captures both dominant common routines and diverse fine-grained behaviors. The appendix section *Cross-city Analysis* further confirms the potential for large-scale, real-world deployment across diverse urban environments. The representative Shanghai description in Fig. 7 and Nanchang case in Appendix Fig. 13, shows that both cities share a post-lunch transition from entertainment usage to utility or work-related applications, while differing in the specific transition time. This indicates that the collective knowledge base preserves shared behavioral regularities as well as city-specific temporal characteristics. Additional cases are provided in the appendix section *Collective Knowledge Analysis*, where collective knowledge consistently improves representative deep learning models and demonstrates strong reusability.

Reflection Effect Analysis We evaluate the effect of the reflection across prediction iterations. As shown in Fig. 6, reflection improves performance as iterations proceed, demon-

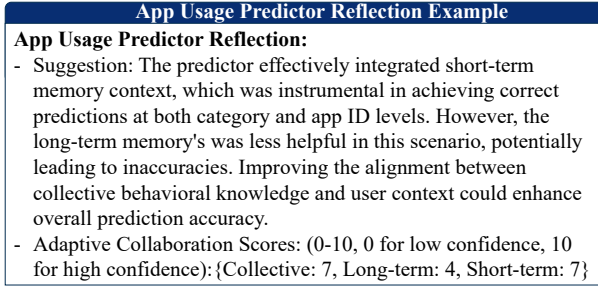


Figure 5: A representative reflection case on the app usage predictor, showing how reflection refines agent behavior and adapts collaboration among agents.

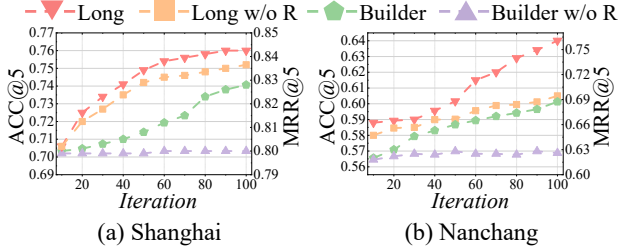


Figure 6: Effect of the reflection mechanism on the collective pattern builder and the long-term personalized pattern profiler. The horizontal axis denotes prediction iterations, and the vertical axis denotes app prediction ACC@5.

strating that reflection signals help refine collective behavior semantics, update personalized memory summaries, and dynamically calibrate the predictor’s reliance on different contextual sources. Fig. 5 presents a representative reflection case for the app usage predictor, where the reflection signal evaluates contextual contributions and assigns adaptive collaboration scores. This case shows that reflection enables the predictor to correct reliance on less informative contexts. Additional analyses and cases are provided in the appendix section *Reflection Effect Analysis*.

LLM Backbone We replace the default LLM with representative models while keeping the data, prompts, agent roles, and hyperparameters fixed. As shown in Fig. 10, stronger backbones improve performance, with GPT-4o performing best overall. This indicates that semantic reasoning benefits agent collaboration; detailed comparisons are provided in the appendix section *Impact of LLM Backbone*.

Communication Overhead Analysis We analyze the inference latency and communication overhead introduced during the multi-agent reasoning process and interactions. We measure the average inference latency, communication overhead, and token consumption per app prediction and reflection on the Shanghai dataset, and analyze the impact of reflection frequency on them. As shown in Table 3, we observe that the majority of latency comes from the reflection, as it involves additional reasoning processes, including updating

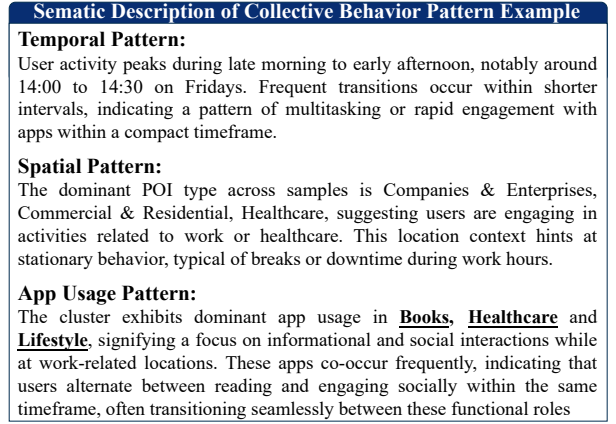


Figure 7: An example semantic description of a popular behavior group in Shanghai.

Table 3: Average inference latency, communication overhead, and token consumption per app prediction and reflection on the Shanghai dataset.

Cost	inference latency	communication overhead	input token consumption	output token consumption
prediction	2.12±0.60s	0.32±0.12s	342.76±51.09	85.30±12.75
reflection	5.20±1.03s	0.96±0.14s	623.22±71.86	249.01±15.13
total	7.32±1.63s	1.28±0.26s	965.98±122.95	334.31±27.88

the knowledge base and refining personalized memory, which introduce extra computational and communication overhead. Moreover, improvements in hardware and system optimization are expected to further reduce the computation time. The appendix section *Communication Overhead Analysis* additionally shows that decreasing the reflection frequency can reduce the cost.

Conclusion

In this paper, we propose AgentApp, a center-user multi-agent framework that bridges personalized and collective behavior patterns to predict mobile app usage through adaptive collaboration. AgentApp comprises a collective pattern builder on the cloud center-side, and a personalized pattern profiler, an app usage predictor, and an adaptive collaboration reflector on each local user-side. Experiments on two real-world datasets demonstrate that AgentApp consistently outperforms baselines. Further evaluations confirm AgentApp’s effectiveness, robustness, transferability, and scalability.

Ethical Statement

All datasets used in this study were collected with proper authorization from users and data providers. The data were anonymized and desensitized before research use, preventing researchers from accessing private user identities or personally identifiable information. In the proposed user-crowd collaborative architecture, individual behavioral records and personalized memories are stored on the user side. Information shared with the cloud-side crowd module is anonymized.

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